6.8 Identical particles in a molecule

Consider a molecule made of identical particles which is rotated by 180 degrees so that the identical particles are exchanged. Under rotation by 180 degrees the wavefunction is altered by a factor of $(-1)^l$ where l is the angular momentum

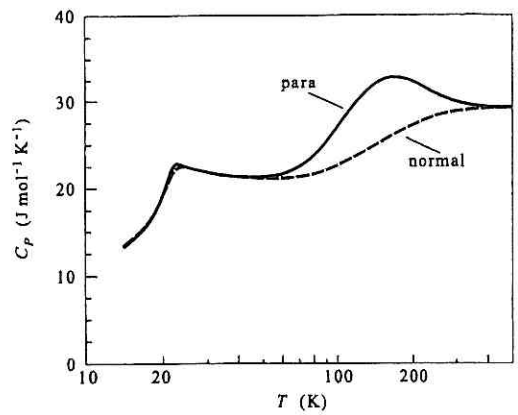

Fig. 6.3 The molar heat capacity of para- and normal hydrogen as a function of temperature from the data tabulated by Vargaftik (1975). Below 50 K the two heat capacities are identical and arise from translational motion; above 50 K the heat capacities are different because the rotational parts of the partition function are not the same: the partition function for para-hydrogen is given by eqn (6.8.1) whereas for normal hydrogen it is given by eqn (6.8.2).

quantum number as discussed in Appendix C. An 's' orbital is unchanged by such a rotation, a 'p' orbital changes sign.

As an example, consider a CO_2 molecule with two ^{16}O nuclei which is rotated by 180 degrees. The ^{16}O nuclei are bosons with spin zero and hence the spin wavefunction is symmetric. The total wavefunction must keep the same sign on rotation by 180 degrees. It follows that only even values of l are allowed by the Pauli exclusion principle. The rotational partition function is

$$Z = \sum_{l=\text{even}}^{\infty} (2l+1)\, e^{-\hbar^2 l(l+1)/2Ik_BT}. \tag{6.8.1}$$

Another example is a hydrogen molecule. The nucleus of each atom is a proton, a fermion of spin 1/2. The total wavefunction of the protons must change sign on rotation by 180 degrees. The spin state of the protons with total spin zero is $\chi_4 = |\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle$ which is antisymmetric on interchange of particles. The orbital wavefunction of the protons must keep the same sign on interchange of particles and so l must be 0, 2, 4, ... , an even integer. This type of hydrogen is called *para-hydrogen*.

The three spin states χ_1, χ_2, and χ_3 form a triplet with a total spin of one. They are all symmetric on interchange of particles. The orbital wavefunction of the protons must change sign on interchange of particles and so l must be 1, 3, 5, ... , an odd integer. This type of hydrogen is called *ortho-hydrogen*.

Ordinary hydrogen consists of a mixture of para- and ortho-hydrogen. The total partition function is

$$\begin{aligned} Z &= \sum_{l=\text{even}}^{\infty} (2l+1)\, e^{-\hbar^2 l(l+1)/2Ik_BT} \\ &\quad + 3\sum_{l=\text{odd}}^{\infty} (2l+1)\, e^{-\hbar^2 l(l+1)/2Ik_BT}. \end{aligned} \tag{6.8.2}$$

The factor of three in the last term in eqn (6.8.2) is for the three spin states. The adjustment to chemical equilibrium of these two species is very slow except in the presence of a catalyst. We can think of them as two different species of molecule each with its own heat capacity.

The data shown in Fig. 6.3 are the heat capacity of normal hydrogen, which consists of an equilibrium mixture of ortho- and para-hydrogen, together with the heat capacity of para-hydrogen.